Topic: Boundary value problems

Question: Rewrite the boundary conditions for the product solution $u(x, t) = v(x)w(t)$.

$$\frac{\partial^2 u}{\partial t^2} = k\frac{\partial u}{\partial x}$$

$$u(x,0) = 0,\ u(0,t) = 0,\ \frac{\partial u}{\partial x}(L, t) = 0$$

Answer choices:

A $w(0) = 0,\ w'(L) = 0,\ v(0) = 0$

B $v(0) = 0,\ v'(0) = 0,\ w(0) = 0$

C $v(0) = 0,\ v'(L) = 0,\ w(0) = 0$

D $v(L) = 0,\ v'(L) = 0,\ w(0) = 0$

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Solution: C

Substituting $u(x,0) = 0$ into the product solution gives

$$u(x, t) = v(x)w(t)$$

$$u(x,0) = v(x)w(0) = 0$$

When $v(x) = 0$ we get the trivial solution, so we'll only use $w(0) = 0$.

Substituting $u(0,t) = 0$ into the product solution gives

$$u(x, t) = v(x)w(t)$$

$$u(0,t) = v(0)w(t) = 0$$

When $w(t) = 0$ we get the trivial solution, so we'll only use $v(0) = 0$.

And if we differentiate the product solution and then substitute the partial derivative condition, we get

$$\frac{\partial u}{\partial x} = v'(x)w(t)$$

$$\frac{\partial u}{\partial x}(L, t) = v'(L)w(t) = 0$$

When $w(t) = 0$ we get the trivial solution, $u(x, t) = 0$, so we'll only use $v'(L) = 0$. So the boundary conditions become $v(0) = 0$, $v'(L) = 0$, and $w(0) = 0$.

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Topic: Boundary value problems

Question: Solve the boundary value problem if $y(0) = 2$ and $y'(0) = 0$.

$$y'' - 9y = 0$$

Answer choices:

A $y = e^{3x} + e^{-3x}$

B $y = e^{\sqrt{3}x} + e^{-\sqrt{3}x}$

C $y = e^{3x} - e^{-3x}$

D $y = e^{\sqrt{3}x} - e^{-\sqrt{3}x}$

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Solution: A

The characteristic equation associated with the differential equation is

$$r^2 - 9 = 0$$

$$r = \pm 3$$

So, the general solution of the differential equation is

$$y(x) = c_1 e^{3x} + c_2 e^{-3x}$$

$$y'(x) = 3c_1 e^{3x} - 3c_2 e^{-3x}$$

Substitute the initial conditions $y(0) = 2$ and $y'(0) = 0$.

$$y(0) = 2 \qquad\qquad y'(0) = 0$$

$$c_1 e^{3(0)} + c_2 e^{-3(0)} = 2 \qquad\qquad 3c_1 e^{3(0)} - 3c_2 e^{-3(0)} = 0$$

$$c_1 + c_2 = 2 \qquad\qquad 3c_1 - 3c_2 = 0$$

Solving the system of equations gives $c_1 = c_2 = 1$, so the solution is

$$y = e^{3x} + e^{-3x}$$

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Topic: Boundary value problems

Question: Solve the boundary value problem given $u(\pi/5,y) = 0$, assuming the separation constant is -25.

$$\frac{\partial^2 u}{\partial x^2} - 5\frac{\partial u}{\partial y} = 0$$

Answer choices:

A $u(x, y) = Ce^{-5y} \cos 5x$

B $u(x, y) = Ce^{-5y} \sin 5x$

C $u(x, y) = Ce^{5y} \sin 5x$

D $u(x, y) = Ce^{5y} \cos 5x$

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Solution: B

Starting with the product solution and its partial derivatives,

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial u}{\partial y} = v(x)w'(y)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(y)$$

we'll plug these into the partial differential equation.

$$v''(x)w(y) - 5v(x)w'(y) = 0$$

Separate variables then set the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)v''(x) = \left(\frac{5}{w(y)}\right)w'(y) = -25$$

Rewrite the boundary conditions.

$$u\left(\frac{\pi}{5}, y\right) = v\left(\frac{\pi}{5}\right)w(y) = 0$$

$$v\left(\frac{\pi}{5}\right) = 0$$

Solving the first equation we found with the separation constant, we get

$$\frac{1}{v}v'' = -25$$

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$$v'' = -25v$$

$$v'' + 25v = 0$$

The associated characteristic equation is

$$r^2 + 25 = 0$$

$$r = \pm 5i$$

Then the solution is

$$v(x) = c_1 \cos 5x + c_2 \sin 5x$$

and substituting the initial condition gives

$$v\left(\frac{\pi}{5}\right) = -c_1 = 0$$

$$v(x) = c_2 \sin 5x$$

Solving the second equation we found with the separation constant, we get

$$\frac{5}{w}w' = -25$$

$$\frac{1}{w}w' = -5$$

$$\int \frac{1}{w}\, dw = -5 \int dy$$

$$\ln|w| = -5y + C$$

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$$w = Ce^{-5y}$$

So the solution is

$$u(x, y) = Ce^{-5y} \sin 5x$$

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